SCOUT PROJECT · TEAM 44 · ADVANCES 1 TO 5

SCOUT’s value stream map

SCOUT seen as a supply chain: the “raw material” is MIT’s incident records and the “product” is an evidence-backed report for the person who uses it. Each process shows what goes in, what comes out, why it exists and what we watch.

Proposed diagram. Advances 4 and 5 are planned and do not exist yet.

Process done · Next process · Inventory: what passes to the next step · Information flow (what the user provides)

SUPPLIER

MIT TRACE

797 real cyberattack incidents with supply-chain consequences, 28 data fields each. File dated 28 September 2026.

ADVANCE 1 — DONE

1. Receive and review

▲ 797 records

KEY FACTS
- 28 fields per incident: who was hit, which link of the chain, which functions, attack and impact types
- No field for root cause or vulnerability (no CVE or CVSS score)
- “Estimated severity” reflects how well a case is documented, not how bad it was (confirmed in Advance 3)

WHY
A recommendation is only as good as its evidence: first learn what TRACE can and cannot support.

WHAT WE WATCH
Fields that could mislead, like severity, are kept as context and never used to rank.

ADVANCES 1 & 2 — DONE

2. Clean and remove duplicates

▲ 784 reference cases

KEY FACTS
- Labels unified into one catalog
- 13 pairs of near-duplicate records (the same incident reported twice)
- 678 cases complete enough to compare: sector, chain link and impacts present

WHY
A duplicated incident would count twice and inflate “what tends to go wrong”.

WHAT WE WATCH
Each incident counts only once in the results.

ADVANCE 2 — DONE

3. Describe every case the same way

▲ Records with checkboxes and a fingerprint

KEY FACTS
- 130 checkboxes: sector, chain link, function, attack, impact, role
- A 384-number “fingerprint” per case; the search uses the fingerprint of the profile, not of the story

WHY
The computer can only find “companies like yours” if past cases and users are described in the same terms.

WHAT WE WATCH
Compare like with like, using the questionnaire’s own options.

ADVANCE 2 — DONE

4. Store in the index

▲ Index ready to search

KEY FACTS
- 3 indices (profile, story, keywords), exact search
- Answers in a fraction of a second over about 800 cases

WHY
A fast, exact lookup lets the app answer while the user waits.

WHAT WE WATCH
The search never skips a case: checked against comparing one by one.

continues below ↓

The SCOUT user’s questionnaire enters here

ADVANCES 2 & 3 — DONE

5. Find the 5 most similar cases

▲ 5 cases, each marked strong or weak

KEY FACTS
- Blend: 60% profile, 40% keywords
- Quality 0.87 on a 0–1 scale (random 0.29; the same list for everyone 0.48)

WHY
The incidents of similar companies are the evidence behind every later recommendation.

WHAT WE WATCH
Each case is marked strong or weak. The mark said “strong” in 93 of 100 tests, so it will be recalibrated with human review.

ADVANCES 2 & 3 — DONE

6. Count and write with citations

▲ Report with citations

KEY FACTS
- The program, not the AI, computes the consequence table (“4 of 5”, and how many times more often than usual)
- The AI must cite a case for every sentence, and automatic checks verify it. Its most frequent errors are counts it writes in its own words

WHY
Every statement must be traceable to a real incident, never invented.

WHAT WE WATCH
With fewer than 2 strong matches it does not write. If a check fails, it shows a template summary instead.

ADVANCE 3 — PARTIAL

7. Quality inspection

▲ The bar every improvement must beat

KEY FACTS
- Tested on companies it had never seen
- Consequences: 0.63 vs 0.20 for giving everyone the most common ones (0.46 with sector alone)
- Search: 0.87 vs 0.48
- AI text: 5 of 8 checks passed (HHEM, an AI judge, counterfactual tests). More useful than the template (0.75 vs 0.64), less faithful (0.59 to 0.79 vs 1.00)

WHY
Before building more: prove it beats the obvious and know its limits.

WHAT WE WATCH
Beating the obvious. Human review is pending; it will calibrate the automatic checks.

ADVANCES 4 & 5 — NEXT

8. Recommend and prioritize

▲ Ranked actions

WHAT WE WILL DO
- Link each action to the attacks and consequences it addresses, reviewed by Dr. Mora
- Priority = similar cases × process criticality × exposure (CVSS v4.0, CISA BOD 26-04)

WHY
Turns “what happened to others” into “what you should do first”.

WHAT WE WILL WATCH
Compare against a simple list of actions and against expert opinion.

CUSTOMER

The SCOUT user

Receives today: similar cases, what tends to go wrong and questions to prepare for, each one cited.

Later: a value stream map of their own company colored by risk, with ranked actions and the evidence behind each.

ADVANCES TIMELINE · WHICH PART OF THE MAP EACH DELIVERY COVERS

Advance 1
Processes 1 and 2: know and clean the data

Advance 2
Processes 2 to 6: describe, index, search and write

Advance 3
Processes 5 to 7: test against the obvious and audit the AI’s text

Advance 4 (plan)
Process 8: action catalog

Advance 5 (plan)
Process 8: priority and comparison of versions

Working now
- Cleaning and checkbox records
- Text fingerprints and search
- Counts and cited writing, with safety nets
- Tests against the obvious; AI-text audits with HHEM and an AI judge

Still to close
- Human review of cases and answers, to calibrate the automatic checks
- Have the program, not the AI, write every count
- A finer usefulness scale for the AI judge
- A questionnaire that adapts to the chain link

Later
- Actions and priority score
- The user’s value stream map with risk colors
- Vulnerabilities (CVSS v4.0, CISA BOD 26-04) if the questionnaire includes them

THREE TERMS IN PLAIN WORDS

RAG: the AI first looks up real cases, then writes only what they support, citing each one. · **Baseline:** the simple reference a smarter method must beat. · **Fingerprint (embedding):** numbers that capture what a text means, so similar cases sit close together.

Figures from Team 44’s Advance 1, 2 and 3 notebooks; Advance 3 figures come from the run with Claude Haiku 5.5 writing and Claude Sonnet 5.5 judging. Advances 4 and 5 follow the plan in the Advance 3 notebook. MIT TRACE cases are real, published incidents.